Endogenous Screening and Health Behaviors*

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Abstract

We study....

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Introduction

1. Preventive care is interesting because it presents an intertemporal tradeoff associated with a foundational form of human capital; it is "underutilized"; and it carries enormous policy implications.
   - There is heterogeneity in who seeks preventive care: https://jamanetwork.com/journals/jamanetworkopen/fullar
2. We think about this in the context of lung cancer screening. It is effective; safe; thought to be underutilized; and, unlike other types of cancer screenings, tied to a specific behavior: smoking.
   - LCS underutilized: https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2816601
3. We build a structural model of screening and smoking that embeds the following mechanisms: Information, disutility, cost, and dynamics.
4. We bring three exciting data sets to this: novel survey, truveta, and BRFSS.
5. Reduced-form results:
   - Survey experiment about screening: information, stigma, cost.
   - We find that individuals are highly sensitive to information about prices with regards to screening decisions. We also find that eliminating stigma and providing information about screening quality improves intent to screen among affected populations. Finally, our experimental evidence indicates that negative screening outcomes indicating no cancer encourage smoking cessation, an important policy-relevant result.
   - RDD about eligibility. This is a useful moment because it says something about the price elasticity of screening and the individual responsiveness to risk.
6. Counterfactual Simulation results
7. Connections to the literature.

- Information
  - About LCS existence
  - About Effectiveness/Screening Quality.
  - About cost.
  - About the benefits.
  - About Lung Cancer itself
- Disutility
  - Stigma
  - Of the procedure
  - Fear of diagnosis
  - Radiation exposure
  - Language and Cultural Barriers
- Cost

  - Financial Cost
  - Time Cost
  - Geographic Accessibility

- Other
  - Lack of Trust in providers/technology
  - Fatalism
  - Value of life effects
  - Competing Risk

Background

In 1996, USPSTF gave lung cancer screening a D rating. Their methodology was updated prior to the 2004 report. The 2004 report gave LCS an I grade. They cited evidence that LCS can detect lung cancer at an earlier stage, but they cite poor evidence on decreases in mortality. They cite poor rate of false positives and invasive screening. Note that invasive procedures refer to tests/diagnostics after a false/positive screening. They cite a lack of RCT evidence on mortality effects.

The National Lung Screening Trial (NLST) Team (2011) funded by NCI was a randomized trial to determine if LDCT screening reduced mortality relative to chest radiography. This was done in high risk persons. It started in 2002, with follow up data through October 2010. The result was a significant decline in both death from LDCT and all-cause mortality. Details:

- Enrollment 8/2002 - 4/2004
- Screening from 8/2002-9/2007.
- Followup through 12/2009.
- population: 55-74 with py history of 30 or more and recent smoking within 15 years.
- They excluded: prior lung cancer, screened with CT in prior 18 months, hemoptoysis, or unexplained weight loss.
- n=53,454, enrollment at 1 of 10 centers
- Treatment included 3 screenings at 1-year intervals. If lung cancer was found, no further screenings.
- at baseline, 48% were former smokers.
- Higher positive test rate in CT arm. 7191/26309 vs. 2387/26035
  - Of those receiving a positive test in round 0, only 3.8% were confirmed to be lung cancer. 5.7% in the chest radiography.
- Main result: 1060 LC diagnosed in LDCT arm; 941 in radiography group; 247 deaths from LC in LDCT group vs. 309 in the radiography group.

The NLST was transformative. In 2013, USPSTF revised their recommendation for LDCT to a B usp (2014). The recommendation was issued on Dec. 31st, 2013. This was for adults 55 to 80 with 30+ pack-years and currently or recently (within 15 years) have smoked. Screening should be stopped after a person has been non-smoker for 15 years. Interesting note: also discontinue screening if they develop a health condition that "substantially limits life expectancy or the ability or willingness to have curative lung surgery."

- 10-12% of screen-detected cancer cases are overdiagnosed, they would not have been detected in the patient's lifetime without screening.
- One thing that these recommendations emphasize is that the B rating of moderate net benefit is for "high risk" individuals, which they explicitly define above. The BRFSS data suggest that many individuals have a different definition of high risk.
- usp (2014) recommend screening through age 80. CMS only pays through age 77, which comes from the fact that the NLST trial enrolled to age 74 at baseline and there was a 3 year treatment.

usp (2014) statement on shared-decision making: "Shared decision making is important for the population for whom screening is recommended. The benefit of screening varies with risk because persons who are at higher risk because of smoking history or other risk factors are more likely to benefit. Screening cannot prevent most lung cancer deaths, and smoking cessation remains essential. Lung cancer screening has substantial harms, most notably the risk for false-positive results and incidental findings that lead to a cascade of testing and treatment that may result in more harms, including the anxiety of living with a lesion that may be cancer. Overdiagnosis of lung cancer and the risks of radiation are real harms, although their magnitude is uncertain. The decision to begin screening should be the result of a thorough discussion of the possible benefits, limitations, and known and uncertain harms."

February 5th, 2015, CMS issues the following National Coverage Decision. CMS can do this under 1861(ddd) of the social security act. Additional preventive services through NCD if reasonable and necessary for prevention or early detection of illness; A or B grade from USPSTF; appropriate for individuals under parts A or B.

- Coverage added under Part B
- LCS counseling and share decision making visit + for appropriate benes, annual screening for lung cancer with LDCT.
- Criteria to be eligible:
  - age 55-77
  - asymptomatic
  - 30 pack years or more
  - current or recent (within 15 years) smoker
  - written order for LDCT LCS that lists:
    - beneficiary date of birth
    - pack year history
    - current smoking status
    - statement that patient is asymptomatic
    - NPI of ordering practitioner.

Before LDCT can take place, the beneficiary must receive a counseling and shared decision making visit that meets the follwoing criteria:

- must be furnished by a physician or qualified non-physician practitioner.

usp (2014)

Of those eligible according to the 2021 guidelines, Bandi et al. (2024) find that 18% are up to date (which I assume means within the last year) on their screening. They describe this as low. Henderson et al. (2024), study the same data and focus on the same population. They find prevalence to be 16.4%. Darden and Hoagland (2025) find that nearly half (45.2%) of past-year screenings came from people outside of USPSTF guidelines and therefore likely paid out-of-pocket.

Timeline of Policy Changes

May 15th 2004

USPSTF: evidence is insufficient to recommend for or against screening. Grade I

August 4th 2011

"Screening with the use of low-dose CT reduces mortality from lung cancer" (Team, 2011).

December 31st 2013

USPSTF: Recommended CT screening in adults aged 55 to 80 with 30 pack-year history and who currently or recently (<15 years) smoke. Grade B

February 5th 2015

Medicare National Coverage Determination under USPSTF guidelines (55 to 77) + shared decision making visit before first screening.

March 9th 2021

USPSTF: Recommended CT screening in adults aged **50** to 80 with **20** pack-year history and who currently or recently (<15 years) smoke. Grade B

February 10th 2022

Medicare National Coverage Determination under USPSTF guidelines (50 to 77) + shared decision making visit before first screening.

- Age 55-77 years.
- Asymptomatic.
- 30+ pack-year history.

- Current or Recent (<15 years) smoker.
- Written Order.
- Shared decision making visit.

Theory

The goal of the model is to rationalize the observed rate of screening. To do so, we model screening and smoking decisions jointly in an environment with uncertainty about disease state and the quality of screening. We consider two forms of health: lung cancer and other chronic health. Unless lung cancer is advanced such that symptoms are apparent, the state is unobserved by both the individual and the econometrician. Early stage lung cancer, if diagnosed, can potentially be cured, but advanced lung cancer cannot be cured. Other health is a binary variable for the presence of any other chronic health condition, which introduces the notion of competing risk. We present the model in the order of a representative period t.

State Variables

$$\Gamma_t = (\theta_t, d_t, b_t, h_t, \phi_t, \nu_i, a_t)$$

- $\theta_t \in \{1, 2, 3\}$: Lung cancer state corresponding to none, early-stage, and late-stage lung cancer. θ is observed only when detected via screening (in the absence of a false negative screen).
- $d_t \in \{0, 1\}$: diagnosis state, where $d_t = 1$ indicates that an individual has detected their early- or late-stage cancer.
- b_t: the subjective probability of any lung cancer. If $d_t = 1$, then $b_t = 1$. We abstract away from separate belief distributions of early- and late-stage lung cancer by using the fact that roughly 50% of new lung cancer diagnoses in the U.S. are late-stage.
- h_t: Smoking history, measured in pack-years.
- $\phi_t \in \{1, 2\}$: Other health state. If $\phi_t = 2$, this indicates that an individual has chronic conditions that are highly co-morbid with lung cancer, including COPD and cardiovascular conditions.
- $\nu_i \in \{1, 2, 3, 4\}$: Individual type. Individuals have either high or low stigma associated with testing, and high or low optimism about test quality (based on their subjective beliefs about screening quality relative to the average below). Hence, there are 4 possible types.

There is also exogenous age a_t. In this model, age and time are synonymous. We consider a time horizon from age 40 to age 100, at which point death occurs with probability 1.

Start of the Period: Symptoms

At the start of a period (year), an individual receives a draw of symptoms influencing both their beliefs about their lung cancer risk and their utility. In each period, let η_t represent this draw, where:

$$\eta_t \sim F(\eta|\theta_t, \phi_t).$$

For tractability, we assume that symptoms are normally distributed, with mean and variance determined by the true underlying stage of cancer (including no cancer) and the presence of other health conditions. Based on this draw

of symptoms, an individual's period-specific beliefs about any cancer (where conditional beliefs about early- and late-stage cancer are approximately equally distributed) are given by a conjugate posterior. In practice, we calibrate these symptom draws to match prior literature, such that the probability of observing symptoms is 0.033 if $\theta = 1$, 0.064 if $\theta = 2$, and 0.246 if $\theta = 3$. Given $\eta = 1$, the resulting belief updates can be expressed as:

$$\tilde{b}_t|\eta = 1 : \frac{0.156\, b_t}{(0.156\, b_t + 0.033\,(1 - b_t)}$$
$$\tilde{b}_t|\eta = 0 : \frac{(1 - 0.156)\, b_t}{((1 - 0.156)\, b_t + (1 - 0.033)\,(1 - b_t)}.$$

Actions

Based on their state, an individual simultaneously chooses whether to smoke cigarettes and whether to screen for lung cancer.

$$A_t = (s_t, x_t)$$

- $s_t \in \{0, 1\}$: Smoking decision
- $x_t \in \{0, 1\}$: Screening decision. Note that if $\theta_t = 3$ is known by the individual, no screening occurs ($\theta = 3$ is an absorbing state).

Screening Outcomes

At the time an agent makes smoking and screening decisions, there are two outcomes that are uncertain, both of which are observed if the agent chooses to screen. First, conditional on screening, they receive a signal of early stage lung cancer $z \in \{0, 1\}$. The quality of the test depends on its rates of false positives and false negatives. Define:

$$\Pr(z_t = 1 \mid \theta_t) = \begin{cases} \rho & \text{if } \theta_t = 2 \\ \lambda & \text{if } \theta_t = 1. \end{cases}$$

In this case, ρ is the accuracy of the test (i.e., sensitivity), the probability that the test reveals a true early-stage lung cancer. Thus, $1 - \rho$ is the probability of a false negative. Similarly, λ is the probability of a positive test result in the absence of early-stage lung cancer (i.e., the rate of false positive), and $1 - \lambda$ is the probability of a correctly negative test result. Both ρ and λ have objective average values that we intend to estimate from Truveta data. However, what matters for behavior are an individual's subjective beliefs about $\hat{\rho}$ and $\hat{\lambda}$. We intend to measure average subjective beliefs from the survey.

Treatment is the second outcome. In the event that an agent receives a positive signal, we assume that the agent undergoes additional testing. The subsequent testing, which we assume incurs the same cost as the initial testing, is able to precisely determine the value of $\theta \in \{1, 2, 3\}$. If $\theta_t \in 2, 3$, we assume that the agent undergoes immediate treatment y, which affects the probability that early-stage θ will transition to late-stage cancer at the end of the period. If the screening signal is positive (i.e., if $z_t = 1$) and the agent is free of lung cancer (i.e., $\theta_t = 1$), then we assume that subsequent testing, which is still costly, rules out lung cancer.

Utility Function

Utility is a function of the current state, the actions selected, and the outcomes of screening, if screening is chosen. Specifically, we specify a CRRA function of income net of expenditures on screening, smoking, and cancer treatment, and additively separable terms for smoking, screening, and treatment. Utility from smoking depends on its price p_s and the history of smoking h_t, which captures reinforcement (Becker and Murphy, 1988). The financial cost of screening is given by p_x, and the disutility of screening (including stigma) is given by γ_4.

$$
\begin{aligned}
u(s_t, x_t; \Gamma_t) = & U_\theta + \frac{(W_t - p_s s_t - p_x x_t - p_y y_t)^{1-\gamma_0}}{1-\gamma_0} \\
& + (\gamma_1 + \gamma_2 h_t + \gamma_3 h_t^2) \cdot s_t \\
& - \sum_{v \in \nu} \gamma_{4,\nu}\, x_t\, \mathbb{1}\{\nu_i = v\} - \gamma_5 \tilde{p}_t - \gamma_6 \tilde{b}_t - \gamma_7 \mathbb{1}\{\eta_t > 0\}.
\end{aligned}
$$

Utility is health state specific in the sense that we normalize the utility of death to 0. To ensure positive utility when alive, we also specify an additive parameter U_θ that we allow to vary by whether an individual has late-stage cancer ($\theta = 3$) or not ($\theta < 3$). To the extent that $U_{\theta<3} > U_{\theta=3}$, these utility shift terms create dynamic incentives for good health, and they are identified by individual's willingness to engage in preventive care. The second term represents CRRA utility over residual consumption, where wealth W_t is net of expenditures on smoking, screening, and treatment based on their respective prices; γ_0 is the typical CRRA risk aversion coefficient. Smoking may directly affect utility (γ_1), as well as through linear and quadratic reinforcement effects through smoking history (γ_2, γ_3). The summation term introduces type-specific heterogeneity in the marginal disutility of screening, with $\gamma_{4,\nu}$ indexing differential costs by individual type. $\gamma_5 \tilde{p}_t$ captures utility loss from realized OOP medical spending, where $\tilde{p}_t$ is the expected price of both the initial test (which may be free to patients who qualify for screening under the USPSTF guidelines) and any confirmatory testing. Finally, γ_6 captures disutility associated with beliefs about lung cancer risk (affecting incentives to quit smoking and seek treatment) and γ_7, disutility from symptoms in each period.

If $d_t = 1$, the individual is aware that they have cancer. If cancer is early-stage, we assume that screening is automatic in order to manage stage progression; if $\theta = 3$, we assume no screening is required. If, on the other hand, $d_t = 0$, the agent is unaware of the value of θ, and this generates uncertainty about the payoff from screening. Thus, the contribution of current utility to the value function depends on the diagnosis state and screening choice. If there is no screening, then the contribution of current utility is simply $u(\cdot)$; with screening, however, expected current utility is the expectation over the likelihood of θ and the (symptom-updated) beliefs about the quality of the screening test:

$$
\begin{aligned}
E(u(x_t, s_t; \Gamma_t)) = (1 - \tilde{b}_t)\, & [(1-\lambda)\, u(x_t, s_t; z_t = 0, \theta_t = 1) + \lambda\, u(x_t, s_t; z_t = 1, \theta_t = 1)] + \\
& + \tilde{b}_t\, [(1-\rho)\, u(x_t, s_t; z_t = 0, \theta_t \geq 2) + \rho\, u(x_t, s_t; z_t = 1, \theta_t \geq 2)]\,.
\end{aligned}
$$

This term is the weighted average of utility over the two possible cancer states (collapsing early- and late-stage cancers) and the signals from symptoms and screening, with weights determined by subjective probabilities both about underlying cancer incidence ($\tilde{b}_t$) and screening quality (ρ and λ).

Screening Price

The price of screening depends on the agent's state and on exogenous health insurance and location. Because the USPSTF guidelines dictate insurance coverage, we assume that the out-of-pocket price of screening is \$0 if an individual has health insurance and meets the 2021 criteria. We also model geographic access, allowing the full price of screening to reflect the travel and opportunity costs for agents who do not live in an area with screening. Specifically, let

$$
p_x = 1[ineligible] * p + 1[distant] * w
$$

where eligibility requires that $h_t \geq 20$ and $age \in [50, 79]$. The travel distance, which we capture with a simple binary variable, times the wage rate represent the opportunity costs of screening.

State Evolution

Age and Smoking History
There is no uncertainty about an individual's smoking history. Define the evolution of deterministic state variables smoking history and age:

$$h_{t+1} = h_t + s_t, \quad a_{t+1} = a_t + 1.$$

Cancer Transition
We make simplifying assumptions that the probability of late stage lung cancer conditional on no lung cancer is zero ($Pr(\theta_{t+1} = 3|\theta_t = 1) = 0$). We also, given the structure of our utility function, ignore recovery from early-stage lung cancer to "no cancer" and focus only on cancer progression ($Pr(\theta_{t+1} = 1|\theta_t = 2) = Pr(\theta_{t+1} = 2|\theta_t = 3) = Pr(\theta_{t+1} = 1|\theta_t = 3) = 0$). We calibrate parameters governing the following health transitions:

- From $\theta_t = 1$:

$$\pi_{12}(h_t, s_t, a_t) = \Pr(\theta_{t+1} = 2 \mid \theta_t = 1, s_t, h_t, a_t) = \text{logit}^{-1}(\alpha_0 + \alpha_1 h_t + \alpha_2 s_t + \alpha_3 a_t) \tag{1}$$

- From $\theta_t = 2$:

$$Progression: \ \pi_{23}(h_t, s_t, a_t) = \text{logit}^{-1}(\zeta_0 + \zeta_1 \tau_t + \zeta_2 y_t + \zeta_3(\tau_t \times y_t)) \tag{2}$$

$$Stasis: \ \pi_{22}(h_t, s_t, a_t) = 1 - \pi_{10} - \pi_{12}, \tag{3}$$

where τ_t represents the years since diagnosis ($\tau_t = t - T_i^{d_t=1}$).

Cancer Belief Updating
Beliefs at time t ($\tilde{b}_t$) reflect subjective beliefs about any lung cancer at time t (θ_t). The goal is to understand how beliefs evolve given possible screening and cancer outcomes. As cancer may progress even without being detected, beliefs about lung cancer are relevant until cancer is diagnosed ($d_t = 1$). That is, if $d_t = 1$, then beliefs are fixed at 1; otherwise, beliefs update differently given the screening decision and the results of screening. If the agent chooses not to screen, then the posterior belief of any lung cancer is given as:

$$b_{t+1} = \tilde{b}_t + (1 - \tilde{b}_t)\pi_{12},$$

where π_{01} represents the objective transition probability of cancer onset. This says that agents have rational expectations regarding the average evolution of cancer risk. In the case in which they choose to receive no private signal of information, agent's update beliefs based on this average.

If the person chooses to screen and observes a negative signal $z_t = 0$, then uncertainty about θ_{t+1} stems from both the possibility of a false negative test and the average likelihood of transiting to (or remaining in) early stage cancer. Taking these sources of uncertainty in steps, first, the agent updates beliefs from the negative screening test via Bayes' rule: In this case, beliefs are:

$$\tilde{\tilde{b}}_t = \frac{(1-\rho)\tilde{b}_t}{(1-\rho)\tilde{b}_t + (1-\lambda)(1-\tilde{b}_t)},$$

and now accounting for the fact that cancer may develop by $t+1$:

$$b_{t+1} = \tilde{\tilde{b}}_t + (1 - \tilde{\tilde{b}}_t)\pi_{12}.$$

If the person choose to screen and observes $z_t = 1$, then screening has the effect of resolving uncertainty about period t θ, because even a false positive signal generates confirmatory testing. Thus, updated beliefs about period $t+1$ θ are the rational expectations of early stage lung cancer in $t+1$ conditional on the known value of θ_t

$$b_{t+1} = \begin{cases} \pi_{12} & \text{if } \theta_t = 1 \\ 1 & \text{if } \theta_t \geq 2. \end{cases}$$

Health Transitions
We also assume that the other health state $\phi = 2$ is an absorbing state such that:

$$\Pr(\phi_{t+1} = 2 \mid \phi_t = 2) = 1,$$

and

$$\Pr(\phi_{t+1} = 2 \mid \phi_t = 1, h_t, a_t, s_t) = \text{logit}^{-1}(\psi_0 + \psi_1 s_t + \psi_2 a_t + \psi_3(s_t \times a_t)) \tag{4}$$

Mortality

The model also incorporates mortality, the probability of which is increasing in lung cancer and other health states. Note that mortality is not a function of smoking or smoking history—these terms work through lung cancer and other health.

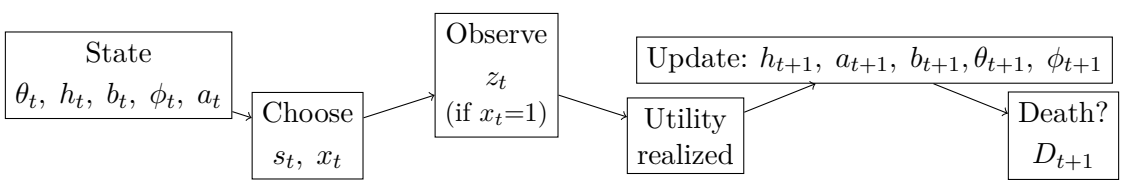
$$\Pr(D_{t+1} = 1 \mid \theta_{t+1}, \phi_{t+1}, a_t) = \text{logit}^{-1}(\omega_0 + \omega_1 \cdot 1[\theta = 2] + \omega_2 \cdot 1[\theta = 3] + \omega_3 \cdot \phi + \omega_4 \cdot a_t) \tag{5}$$

Bellman Equation

Given uncertainty about θ, the value function is the maximum over action space A of the expected value over the perceived distribution of θ.

$$\begin{aligned} V(\Gamma_t) = \max_{A_t} \Big\{ & E(u(s_t, x_t; \Gamma_t)) + \beta \mathbb{E}_{\theta,\phi} \left[(1 - P(D_{t+1} = 1)) V(\Gamma_{t+1}) \right] \\ & + 1[\theta_t = 2] \left[u(s_t, x_t; \Gamma_t, \theta_t = 2) + \beta \mathbb{E} \left[(1 - P(D_{t+1} = 1)) V(\Gamma_{t+1}) \right] \right] \Big\} \end{aligned} \tag{6}$$

Model Summary



Data

BRFSS

1. Relationship between smoking and pack-year history
2. Multiple Dimension RDD of eligibility on screening. This tells us something about selection.u

Survey

We begin by asking standard assessments of risk preferences (Falk et al., 2018) and time preferences (Barsky et al., 1997) that group individual preferences. Next, we assess health insurance status, income, and zip code. Geographic information will be merged at the zip code level to American College of Radiology data lung cancer screening sites.

Next, we assess USPSTF eligibility for subsidized (i.e., $0 OOP) screening. We ask respondents their age, their subjective pack-years of smoking history, and their current smoking status. We also assess beliefs regarding early lung cancer for a generic person that matches their age, smoking history, and overall health. We follow this question

with an assessment of how different the respondent feels they are from average. The goal of these questions is to provide estimates for the learning process in the structural model. Next, we ask respondents about their history with and beliefs regarding LDCT lung cancer screening. These beliefs include subjective assessments of false and true positive rates. We also ask two questions about perceived stigma, both regarding support from family and about the signal that screening sends about past smoking behavior. Together with basic demographic information provided by Prolific, our baseline information on respondents includes measures of smoking and screening history, beliefs regarding lung cancer and screening outcome probabilities, and perceptions of stigma related to screening.

The goal of our randomized information experiment is to understand how preferences, beliefs, perceptions regarding screening quality, stigma, and prices shape the demand for lung cancer screening. The flow chart below depicts the structure of our experimental design. We ask all respondents on a 0 to 100 scale how likely they are to seek screening in the next year. We follow this question with a binary question for screening in the next year. Respondents are randomized into one of four framings of the intention to screen questions. All respondents are eligible to be randomized into framings that are neutral, or that emphasizes screening quality or lack of judgment. For those eligible for free screening, the "price" framing emphasizes their eligibility; for those ineligible for free screening, the price framing includes a randomized OOP price.

All respondents are eligible for a neutral framing of the screening question (control):

> On a scale from 0 to 100, where 0 = definitely will not and 100 = definitely will, how likely are you to seek a lung cancer screening (low-dose CT) in the next 12 months?

Similarly, all respondents are eligible for framings that deal with screening quality:

> Out of every 100 people who get today's standard lung cancer screening (low-dose CT), about 12 will have a false alarm (a positive result when no cancer is present). Most of these are cleared with a repeat scan. On a scale from 0 to 100, where 0 = definitely will not and 100 = definitely will, how likely are you to seek a lung cancer screening (low-dose CT) in the next 12 months?

and a positive framing regarding stigma:

> Lung cancer screening is a clinical test like any other. Whether you smoke now or used to, staff are trained to treat every patient with dignity, and screening results are kept private. On a scale from 0 to 100, where 0 = definitely will not and 100 = definitely will, how likely are you to seek a lung cancer screening (low-dose CT) in the next 12 months?

For those who are eligible and who are randomized to the price treatment arm, they receive the following framing:

> Because of your age and smoking history, the United States Preventive Services Task Force recommends annual lung cancer screening. For most people in your situation, insurance covers the test at no cost to you. On a scale from 0 to 100, where 0 = definitely will not and 100 = definitely will, how likely are you to seek a lung cancer screening (low-dose CT) in the next 12 months?

For those ineligible for screening and who are randomized to the price treatment arm, we frame the question as:

> Because of your age and smoking history, you are not recommended for lung cancer screening under the United States Preventive Services Task Force guidelines. If you choose to receive a screening, most people in your situation pay an out-of-pocket price of about $X. On a scale from 0 to 100, where 0 = definitely will not and 100 = definitely will, how likely are you to seek a lung cancer screening (low-dose CT) in the next 12 months?

We randomly vary the value $\$X \in \{100, 300, 500\}$.

After asking the randomly framed screening intention questions, we ask a series of comprehension questions about the screening that target understanding about the quality of screening, the process of screening (to target

stigma) and the cost. We hypothesize that those receiving the respective treatments should do significantly better on these questions. Next, we assess heterogeneity in differences in intention to screen by treatment arm. Specifically, we hypothesize larger effects of the positive stigma framing among those who, at baseline admit to feeling stigma. Our quality framing presents the true false positive rate of screening; for respondents whose subjective assessment of the false positive rate is higher, our treatment represents "good" quality news, and we expect that our quality treatment should encourage screening. Finally, we assess heterogeneity in the price effects by income.

Our survey ends with a randomized framing of a question regarding expecting smoking behavior over the following 12 months. The control arm receives the following neutral framing:

> Thinking about the next 12 months, please indicate which option best reflects your expectations regarding your smoking behavior.

In contrast, to capture how smoking behavior may respond to the results of lung cancer screening, some respondents are presented with the following framing:

> Suppose a recent lung cancer screening revealed no evidence of cancer. Thinking about the next 12 months, please indicate which option best reflects your expectations regarding your smoking behavior:

Truveta

Empirical Implementation

Estimation

Some parameters, such as those governing health transitions, are calibrated in our model. These parameters are summarized in Table 2. In addition, in our preferred specification of the model, we assume rational expectations, so that beliefs about lung cancer risk are centered around the truth, and fix the coefficient of risk aversion, γ_0, to be equal to 2 in all states. This is done to XXX

We first solve the model by backward induction given an initial guess of all relevant parameters, Θ_0. Solving the model in this way yields estimates of the optimal smoking and screening decisions across the entire state space, meaning for all agents conditional on their demographics, smoking history, and health.

We then identify the model parameters using GMM. For a given solution to the model derived from a parameter guess (such as Θ_0), we simulate smoking and screening behavior for a representative sample of 25,000 agents drawn according to the BRFSS and Prolific data. We then allow these agents' health, beliefs, and utilities to evolve over time according to the optimal policy functions derived above. This allows us to calculate the moments we match on to identify Θ^*.

We use the following moments in estimation:

- Screening rates by age groups among smokers and non-smokers (drawn from our JAMA Network Open paper)
-

Results

Table 3 presents the results from estimating the structural model.

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1 Tables

Table 1: Baseline Covariate Balance

	N	Mean	p_1	p_2	p_3
Age in Years	3912	52.298	0.553	0.769	0.526
Female	3912	0.622	0.571	0.466	0.296
Race					
White	3912	0.825	0.669	0.955	0.336
Black	3912	0.073	0.395	0.705	0.475
Other	3912	0.103	0.924	0.596	0.553
Least Risk Averse	3880	0.053	0.705	0.576	0.428
Most Risk Averse	3880	0.290	0.626	0.671	0.022
Least Patient	3875	0.377	0.535	0.783	0.435
Most Patient	3875	0.086	0.814	0.882	0.979
Uninsured.	3912	0.080	0.955	0.733	0.269
Income					
Less than $25,000	3908	0.161	0.632	0.712	0.194
$25,000-$49,999	3908	0.241	0.570	0.549	0.176
$50,000-$99,999	3908	0.349	0.311	0.922	0.831
$100,000-$199,999	3908	0.209	0.899	0.827	0.555
More than $200,000	3908	0.040	0.765	0.578	0.222
Currently Smoke Cigarettes	3912	0.343	0.233	0.785	0.725
Pack-Years	3912	27.779	0.846	0.396	0.754
Eligibility for Free Screening	3912	0.242	0.714	.	0.593
Age Eligible	3912	0.527	0.320	0.844	0.173
Pack-Years Eligible	3912	0.530	0.804	0.070	0.601
Smoking Eligible	3912	0.710	0.375	0.946	0.451
Ever Screened	3910	0.274	0.271	0.988	0.411
Belief of Early Lung Cancer	3851	22.170	0.231	0.981	0.144
Lower Expected Risk	3912	0.234	0.744	0.216	0.966
About the Same	3912	0.533	0.342	0.070	0.671
Higher Expected Risk	3912	0.233	0.038	0.591	0.647
Belief of True Positive	3851	64.368	0.762	0.466	0.864
Belief of False Positive	3827	17.901	0.849	0.990	0.859
False Positive Information Represents Good News	3827	0.454	0.708	0.563	0.960
High Stigma	3909	0.453	0.031	0.531	0.036
Avoids Information	3912	0.171	0.768	0.162	0.840

Notes: Table 1 presents the overall mean of key variables and three p-values. The first p-value is of the F-test that there are no differences in means between the control arm and the three treatment arms quality, stigma, and price. The second p-value is on the F-test of no differences in means between the lowest price treatment ($100) and the other two prices ($300 and $500) conditional on being randomized to the price arm and being ineligible for free screening. The final p-value is on the t-test that there is no difference in means between the treatment and control arm of the smoking experiment.

Table 2: Calibrated Parameters Governing Health Transitions

Parameters	(1) LC Incidence	(2) LC Progression	(3) Other Health Condition	(4) Mortality
Age	-0.11	1.04	1.32	0.81
Smoking History	0.82	0.07	0.04	–
Currently Smoking	-0.58	0.88	0.24	–
LC Treatment	–	-5.19	–	–
Early-Stage LC	–	–	–	2.35
Late-Stage LC	–	–	–	4.19
Other Health Condition	–	–	–	0.69
Intercept	-8.18	-12.93	-7.68	-6.91

Notes: This table presents calibrated parameters governing how underlying health states evolve over time in the model. These parameters are reported as the coefficients on logistic regressions predicting each of the outcomes indicated in columns (1) through (4). Smoking history is measured in 10 pack-year units, and age is measured in decades above age 40. Parameters are calibrated to match empirical incidence of cancer and its progression based on the NLST, SEER cancer incidence data, and previous work (Darden et al., 2018).

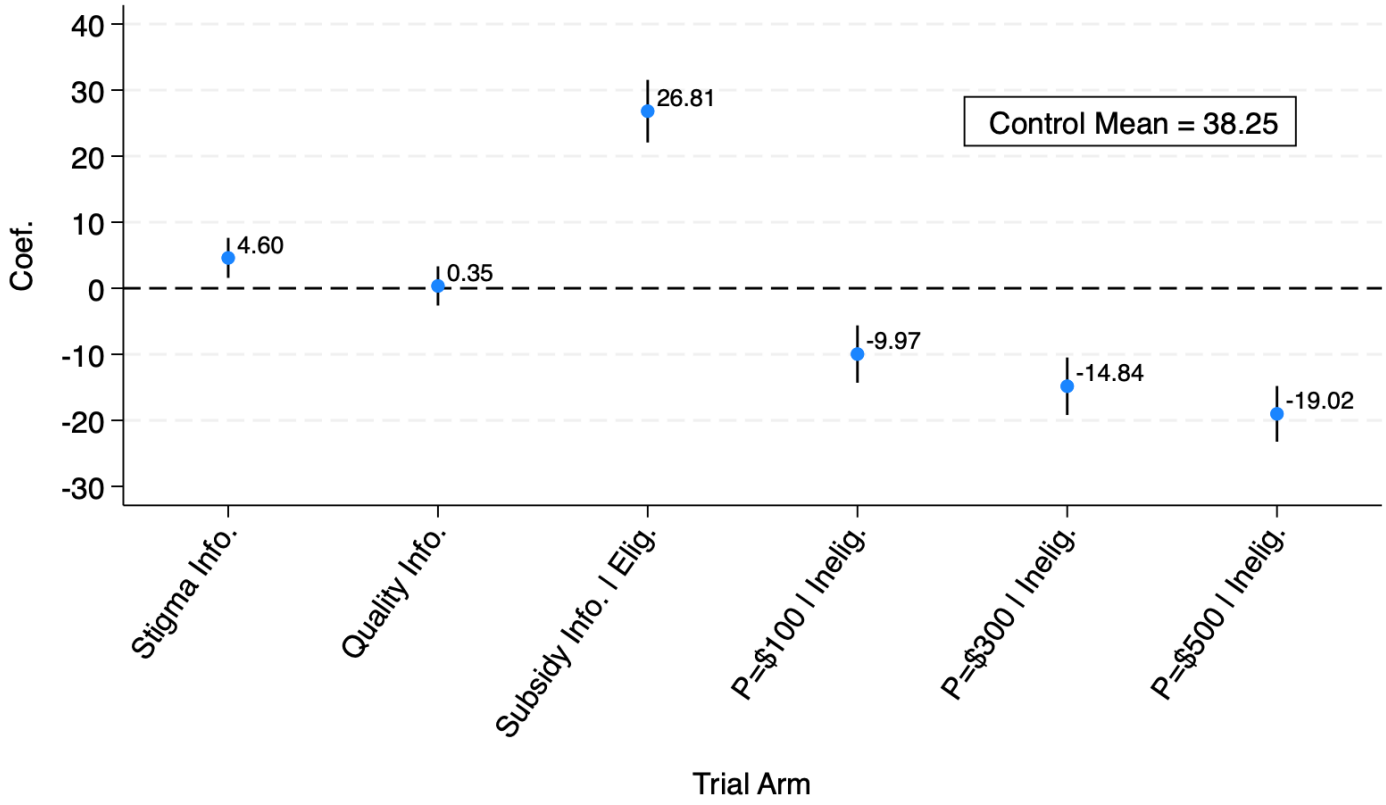
Table 3: Estimated Model Parameters

Parameters	Symbol	(1)	(2)	(3)
Panel A: Utility Function, No Late Stage Cancer ($\theta < 3$)				
Intercept	U_θ		80.81	
		()	()	()
Smoking	γ_1		3,778.85	
		()	()	()
Smoking × history	γ_2		-139.92	
		()	()	()
Smoking × history^2	γ_3		-3.12	
		()	()	()
Screening disutility	γ_4	–	-1999.99	
note: want to incorporate stigma + optimism here?			()	()
Utility loss, OOP	γ_5	–	-8.68	
			()	()
Disutility from high cancer beliefs	γ_6	–		
			()	()
Panel B: Utility Function, Late Stage Cancer ($\theta = 3$)				
Intercept	U_θ		180.15	
		()	()	()
Smoking	γ_1		10,258.00	
		()	()	()
Smoking × history	γ_2		135.63	
		()	()	()
Smoking × history^2	γ_3		-5.50	
		()	()	()
Panel C: Other Parameters				
Initial beliefs, average	$\overline{b_0}$	0.10	0.10	
		–	–	()
Initial beliefs, SD	$\mathrm{sd}(b_0)$	0.04	0.04	
		–	–	()

Notes: This table presents estimated model parameters following Section . Model is estimated via two-step GMM using stochastic gradient descent. [Standard errors forthcoming]

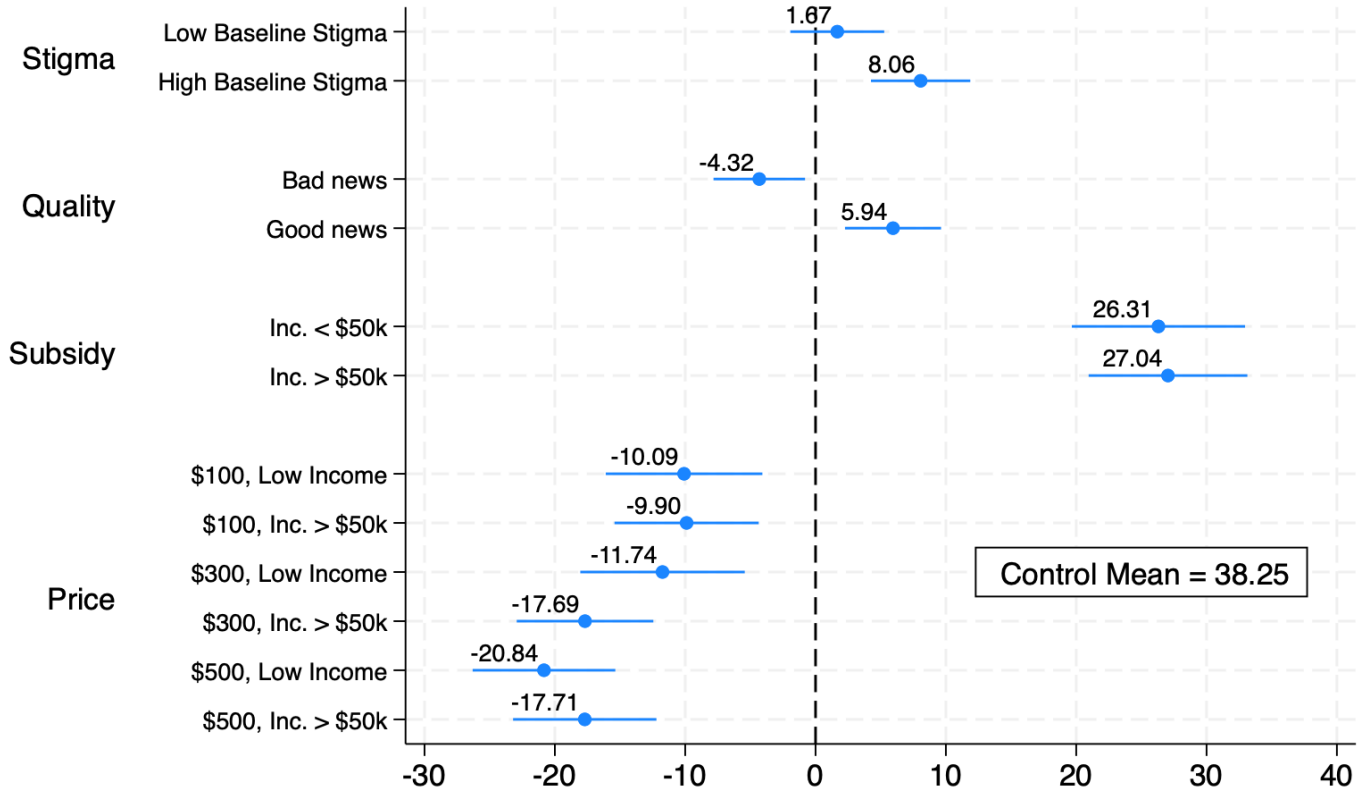
2 Figures

Figure 1: Experimental Effects on Intent to Screen



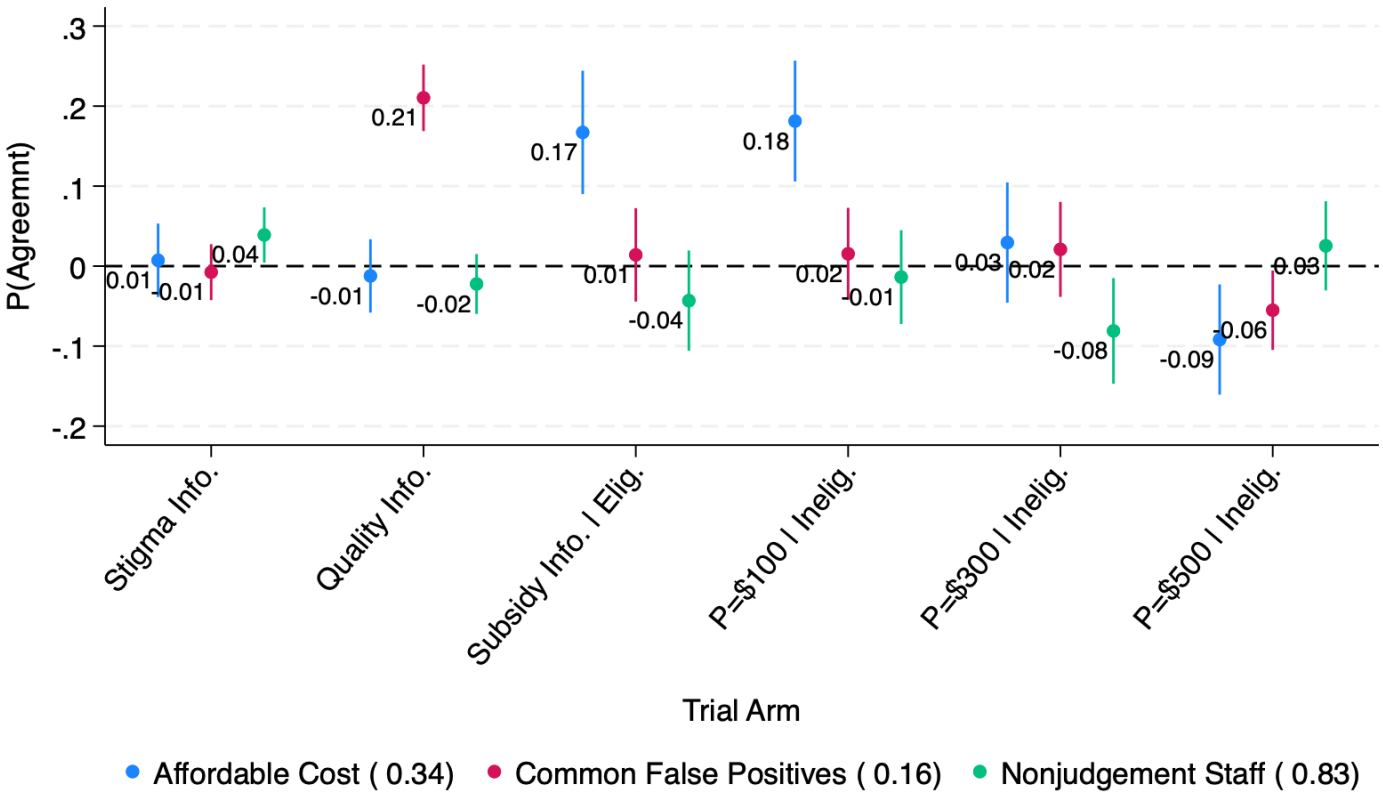
Notes: The Figure presents experimental effects from regressions of the intention to screen in the next year on binary variables for each treatment arm relative to the baseline, neutral framing of the intention to screen question. Treatment arms include receiving information about nonjudgmental screening staff (stigma), the observed false positive rate (quality info) and prices. If randomized to the price arm, eligible respondents received information about their eligibility; ineligible respondents received a price quote between $100 and $500. Confidence intervals reflect robust standard errors. $n = 3,268$.

Figure 2: Heterogeneous Effects on Intent to Screen



Notes: The Figure presents experimental effects from regressions of the intention to screen in the next year on binary variables for each treatment arm relative to the baseline, neutral framing of the intention to screen question. Each treatment arm indicator is interacted with relevant, baseline heterogeneity. High stigma indicates agreement with at least one of the stigma questions. Good news implies that the quality information treatment presented a lower false positive rate relative to baseline expectations. Price/subsidy effects are allowed to vary by whether baseline household income was less than $50k. Confidence intervals reflect robust standard errors. $n = 3,268$.

Figure 3: Comprehension Checks



Notes: The Figure presents regression coefficients of three comprehension check binary variables on binary variables for the treatment arms. The dependent variable is whether the respondent agrees that a.) screening is affordable, b.) false positive are common, and c.) screening staff are nonjudgmental. Confidence intervals reflect robust standard errors. $n = 3,268$.